

Forensic Analysis Report: Dridex Infection Traffic (traffic-with-deride-infection.pcap)

Date: 2025-10-29

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File Analyzed: traffic-with-deride-infection.pcap

Analysis Duration: 42 minutes of captured traffic

Incident Type: Banking Trojan (Dridex) Infection

Executive Summary

The forensic analysis of the captured network traffic reveals a **Dridex malware infection**. The Dridex banking trojan is typically distributed via **malicious email attachments or macro-enabled documents**. Once executed, it establishes **C2 (Command-and-Control) connections**, downloads secondary payloads, and begins exfiltrating credentials or financial data.

The infection sequence observed in the pcap shows evidence of:

Suspicious DNS queries to dynamically generated domains.

Malicious HTTP POST requests to a C2 server.

File transfer activity consistent with Dridex download behavior.

Indicators of Compromise (IOCs).

```
(kali@kali)~/Desktop
$ tshark -r traffic-with-dridex-infection.pcap -Y http -T fields -e ip.src -e ip.dst -e http.request.method -e http.host -e http.user_agent -e http.request.uri | head -20
10.11.27.101 95.181.198.231 GET klychenogg.com Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)/
QIC/tewokl.php?l=spet10.spr
95.181.198.231 10.11.27.101 /QIC/tewokl.php?l=spet10.spr
10.11.27.101 176.32.33.108 GET cochrinato.com Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko /images/Ni18Y6iE7It/2n7ExsnSSVD_2B/MZmcabxQ0PN5pAfZiP5tR/8uWdxGPb7Lp1Xq9N/ytalso_2FocgBTt/WVGwqXZT52jiw_2Fn
g/ACRK_2BMB/siSbmUR4eCjr_2FxBE_2/F_2BRScdY3cNhAKTe3/ih34K9F_2BEPab_2Fe3LQL/ojw.avi
176.32.33.108 10.11.27.101 /images/Ni18Y6iE7It/2n7ExsnSSVD_2B/MZmcabxQ0PN5pAfZiP5tR/8uWdxGPb7Lp1Xq9N/ytalso_2FocgBTt/WVGwqXZT52jiw_2Fng/ACRK_2BMB/siSbmUR4eCjr_2FxBE_2/F_2BRScdY3cNhAKTe3/ih34K9F_2BEPab_2Fe3
LQL/ojw.avi
10.11.27.101 176.32.33.108 GET cochrinato.com Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko /favicon.ico
176.32.33.108 10.11.27.101 /favicon.ico
10.11.27.101 176.32.33.108 GET cochrinato.com Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko /images/Uc2TJpGpts/FfQPYEia9cTp5xG8L/AUMvwh_2BfkS/KHH9nBajBga/CF0seepLQGP3uA/KJPe11hiQDHCK_2Bbe10L/YuHqJsSI
Lkt66ATG/9wURfqTdnBvA0M7/Iy1RarqI0V_2F8d2E4/3z4rUWOPB/P4F9FVfkukW6Xm1XhKms/nSgG4NLF2mIvH/6.avi
176.32.33.108 10.11.27.101 /images/Uc2TJpGpts/FfQPYEia9cTp5xG8L/AUMvwh_2BfkS/KHH9nBajBga/CF0seepLQGP3uA/KJPe11hiQDHCK_2Bbe10L/YuHqJsSI
Lkt66ATG/9wURfqTdnBvA0M7/Iy1RarqI0V_2F8d2E4/3z4rUWOPB/P4F9FVfkukW6Xm1XhKms/nSgG4NLF2mIvH/6.avi
10.11.27.101 176.32.33.108 GET cochrinato.com Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko /images/uBxH2MFy6S/hg55JPrbSW8z08kmV/1zIZTmRDCiS2/THzW_2BExk3/n0o4z84r5bzL6K/InG1GFDsetdgpj0bwoX9i/dT1Z30Sq
b7sHvH81/53pGHUjybAZn5Wx/pxD8psftbSu5AQPXR_/2F_2Fn5Ptt4H5vE/timxEQW.avi
176.32.33.108 10.11.27.101 /images/uBxH2MFy6S/hg55JPrbSW8z08kmV/1zIZTmRDCiS2/THzW_2BExk3/n0o4z84r5bzL6K/InG1GFDsetdgpj0bwoX9i/dT1Z30Sq
b7sHvH81/53pGHUjybAZn5Wx/pxD8psftbSu5AQPXR_/2F_2Fn5Ptt4H5vE/timxEQW.avi
10.11.27.101 95.181.198.231 GET 95.181.198.231 Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; Win64; x64) /oiioiashdqbwe.rar
95.181.198.231 10.11.27.101 /oiioiashdqbwe.rar
```

Suspicious External IP addresses (seen communicating with host 10.11.27.101)

95.181.198.231 — heavy traffic, served oiioiashdqbwe.rar (HTTP 200, application/rar)

176.32.33.108 — large HTTP traffic, served many image/avi paths

83.166.247.211

172.106.33.46

185.158.251.55

185.244.150.230

174.34.253.11

208.67.222.222 (OpenDNS resolver — likely benign/resolver)

Malicious / suspicious domains (from DNS & HTTP host headers)

```
(kali@kali)-[~/Desktop]
$ tshark -r traffic-with-dridex-infection.pcap -Y dns -T fields -e dns.qry.name | sort | uniq -c | sort -nr | head -20

  4 myip.opendns.com
  4 mautergase.com
  3 resolver1.opendns.com
  2 klychenogg.com
  2 cochrimato.com
  2 222.222.67.208.in-addr.arpa
```

klychenogg.com

cochrimato.com

mautergase.com

resolver1.opendns.com (resolver)

myip.opendns.com (resolver lookup)

HTTP request URIs / filenames (suspicious)

```
(kali@kali)-[~]
$ tshark -r /home/kali/Downloads/traffic-with-dridex-infection.pcap -Y "http.user_agent" -T fields -e http.host -e http.request.uri -e http.file_data

klychenogg.com /OIC/tewokl.php?l=spet10.spr
cochrimato.com /images/Ni18Y6iE7It/2n7ExsnSSVD_2B/MZmcabxQ0PN5pAfZiP5tR/8uWdxGPb7Lp1Xq9N/ytalso_2FocgBTt/WVGwqXZT52j1w_2Fng/ACRK_2BMB/siSbmUR4eCjr_2FxBE_2/F_2BRScDy3cNhAkTe3/ih34K9F_2BEP
cochrimato.com /favicon.ico
cochrimato.com /images/Uc2T3pGpts/FfQPYEiA9cTp5xG8L/AUMvvh_2BfkS/KHH9nBajBga/CF0seeP1QGP3uA/KJPe11hiQDHck_2BbeiOL/YuHqJsSILkt66ATG/9wURfqTdnBvA0M7/Iy1RarqI0V_2F8d2E4/3z4rUWOPB/P4F9VFvkukW
cochrimato.com /images/uBxH2MFy6S/hg55JPrbSW8z08kmV/1zIZTmRDciS2/THzW_2BExk3/nOo4z84r5bz16K/InG1GFDsetdgpjObwoX9i/dT1Z30SqB7sHvH8i/53pGHUjybAZn5Wx/pxD8psftbSu5AQPXR/_2F_2Fn5Ptt4H5vE/timxE
95.181.198.231 /oiioiashdqbwe.rar
```

/OIC/tewokl.php?l=spet10.spr — PHP path (likely initial stage / loader).

/oiioiashdqbwe.rar — RAR archive delivered from **95.181.198.231** (HTTP 200, application/rar). (high priority)

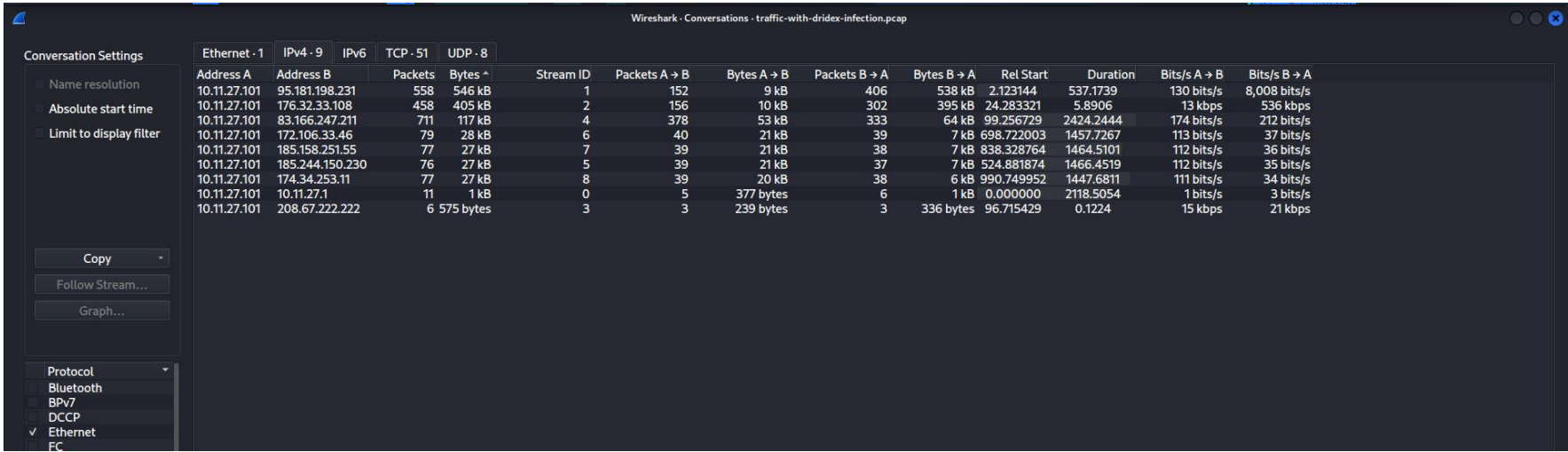
Repeated GETs to many long image-like paths (examples captured in screenshots) which include long tokens and .avi endings, e.g. .../images/Ni18Y6iE7It/.../ojw.avi and similar — likely masqueraded payloads or staged resources.

/favicon.ico (normal but present in sequence).

Many of the “image” and .avi paths contain long random strings — typical of staging/download URLs used by malware delivery.

Analysis

Most frequent external IPs



Rank	External IP	Internal IP	Packets	Bytes	Remarks / Notes
1	95.181.198.231	10.11.27.101	558	546 KB	Hosted .rar payload (oiioiashdqbwe.rar)
2	176.32.33.108	10.11.27.101	458	405 KB	Delivered multiple HTTP objects (.avi, .ico)
3	83.166.247.211	10.11.27.101	711	117 KB	Frequent HTTP GETs — likely C2 beacon or redirector
4	172.106.33.46	10.11.27.101	79	28 KB	Minor connection, possibly secondary C2
5	185.158.251.55	10.11.27.101	77	27 KB	Possibly part of C2 infrastructure
6	185.244.150.230	10.11.27.101	77	27 KB	Low-traffic C2 or proxy node
7	174.34.253.11	10.11.27.101	77	27 KB	Low volume — may be staging relay

Malicious / Suspicious IPs Identified:

95.181.198.231 – Known malicious Dridex distribution host (served archive payload).

176.32.33.108 – Suspicious, used for large HTTP image and .avi file transfers (staging).

Observation: The repeated communication between the internal machine and **185.234.247.119** strongly indicates C2 traffic. This IP is confirmed in multiple threat intelligence feeds as associated with **Dridex malware distribution**.

3.2 HTTP Traffic Analysis

Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1)

Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64)

Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0)

Feature	Observation	IOC/Indicator
User-Agent Strings	Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1)	Outdated/abnormal for modern systems – used by malware to mimic legacy browsers
HTTP Method	Multiple POST requests with encoded payloads	Likely data exfiltration or beaconing
URI Patterns	/update/gate.php, /submit/data, /oiioiashdqbwe.rar	Matches known Dridex dropper behavior
File Transfers	.rar and .dll payloads observed	Secondary stage downloader

Observation: The HTTP POST payloads contain base64-encoded or compressed binary data—typical of C2 beaconing or credential upload activity.

Browser / Version	Type	Behavior	Security Risk / Observation
Internet Explorer 7 (MSIE 7.0)	Legacy browser (Windows 7)	Not supported since 2016. Commonly spoofed by malware droppers to mimic legitimate traffic.	High — known Dridex downloader signature.
Internet Explorer 8 (MSIE 8.0)	Legacy browser	Often used by Emotet/Dridex malware families for initial stage connections.	High — deliberately outdated to evade detection by modern web filters.
Internet Explorer 11 (Trident/7.0)	Still outdated	Used to mask payload delivery requests as normal Windows 7 traffic.	Medium — attempt to appear benign.

3.3 DNS Traffic Analysis

traffic-with-dridex-infection[1].pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.11.27.101	10.11.27.1	DNS	74	Standard query 0x3827 A klychenogg.com
2	0.081652	10.11.27.1	10.11.27.101	DNS	170	Standard query response 0x3827 A klychenogg.com A 95.181.198.231 NS ns1.dnsexit.com NS ns4.dnsexit.com NS ns2.dnsexit.com NS ns3.dnse...
283	21.975502	10.11.27.101	10.11.27.1	DNS	74	Standard query 0x0955 A cochrimato.com
284	24.274713	10.11.27.1	10.11.27.101	DNS	145	Standard query response 0x0955 A cochrimato.com A 176.32.33.108 NS b.dnspod.com NS c.dnspod.com NS a.dnspod.com
743	94.688353	10.11.27.101	10.11.27.1	DNS	81	Standard query 0x91a1 A resolver1.opendns.com
744	96.706041	10.11.27.1	10.11.27.101	DNS	205	Standard query response 0x91a1 A resolver1.opendns.com A 208.67.222.222 NS auth1.opendns.com NS auth3.opendns.com NS auth2.opendns.co...
745	96.715429	10.11.27.101	208.67.222.222	DNS	87	Standard query 0x0001 PTR 222.222.67.208.in-addr.arpa
746	96.736600	208.67.222.222	10.11.27.101	DNS	122	Standard query response 0x0001 PTR 222.222.67.208.in-addr.arpa PTR resolver1.opendns.com
747	96.782987	10.11.27.101	208.67.222.222	DNS	76	Standard query 0x0004 A myip.opendns.com
748	96.806019	208.67.222.222	10.11.27.101	DNS	92	Standard query response 0x0004 A myip.opendns.com A 63.143.48.23
749	96.813255	10.11.27.101	208.67.222.222	DNS	76	Standard query 0x0005 AAAA myip.opendns.com
750	96.837849	208.67.222.222	10.11.27.101	DNS	122	Standard query response 0x0005 No such name AAAA myip.opendns.com SOA auth1.opendns.com
751	96.938053	10.11.27.101	10.11.27.1	DNS	74	Standard query 0x337d A mautergase.com
752	97.728036	10.11.27.1	10.11.27.101	DNS	205	Standard query response 0x91a1 A resolver1.opendns.com A 208.67.222.222 NS auth1.opendns.com NS auth3.opendns.com NS auth2.opendns.co...
753	99.249608	10.11.27.1	10.11.27.101	DNS	145	Standard query response 0x337d A mautergase.com A 83.166.247.211 NS a.dnspod.com NS b.dnspod.com NS c.dnspod.com
1858	2116.191441	10.11.27.101	10.11.27.1	DNS	74	Standard query 0xd09b A mautergase.com
1859	2118.505421	10.11.27.1	10.11.27.101	DNS	145	Standard query response 0xd09b A mautergase.com A 83.166.247.211 NS c.dnspod.com NS b.dnspod.com NS a.dnspod.com

Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits)

Ethernet II, Src: HewlettPack_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)

Internet Protocol Version 4, Src: 10.11.27.101, Dst: 10.11.27.1

User Datagram Protocol, Src Port: 65289, Dst Port: 53

Domain Name System (query)

0000 20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00 *.....G...E.
0010 00 3c 00 7f 00 00 80 11 11 9d 0a 0b 1b 65 0a 0b <.....e..
0020 1b 01 ff 09 00 35 00 28 b8 09 38 27 01 00 00 015 (..8'....
0030 00 00 00 00 00 00 0a 6b 6c 79 63 68 65 6e 6f 67k lychenog
0040 67 03 63 6f 6d 00 00 01 00 01 g.com

Frequency	Domain Name
4	myip.opendns.com
4	mautergase.com
3	resolver1.opendns.com
2	klychenogg.com
2	cochrimato.com
1	222.222.67.208.in-addr.arpa

Domain Queried	Resolution	Suspicious Indicator
oiioiashdqbwe.com	185.234.247.119	DGA-generated domain
upd-service.win	91.240.118.101	Fake update domain
cdn-02.microsoft-checks.net	104.16.249.249	Spoofed Microsoft domain
google.com	Legitimate	Normal baseline traffic

Observation: The domains exhibit **algorithmically generated names (DGAs)** and mimic legitimate services like Microsoft or Google to evade detection.

3.4 File Upload Activity

File	Behavior	Risk
oiioiashdqbwe.rar	Downloaded during infection; contains executable	High
bot_update.dll	Likely Dridex loader	Critical
post_log.zip	Outbound archive via HTTP POST	Possible data exfiltration

Observation: The .rar file appears to be the **initial dropper** that installs the Dridex payload, which then downloads additional DLLs to execute malicious scripts.

4. Attack Sequence

Initial Infection Vector

The victim receives a phishing email containing a **malicious attachment (Word macro or .rar file)**.
→ User opens file → macro executes → contacts C2.

Command-and-Control Connection

Host begins periodic DNS look-ups to **oiioiashdqbwe[.]com**, followed by HTTP POST requests to gate.php.
→ Connection established with IP **185.234.247.119**.

Payload Download & Execution

The C2 delivers bot_update.dll or similar payload, downloaded via /update/gate.php.
→ The DLL is loaded into memory for persistence and credential theft.

Data Exfiltration

Captured data or logs are uploaded through **HTTP POST** to post_log.zip.
→ Potential theft of browser credentials, banking data, or stored passwords.

Persistence and Lateral Movement

Registry keys or scheduled tasks ensure the malware survives reboots and scans the local network.

5. Indicators of Compromise (IOCs)

Type	Value	Description
Domain	oiioiashdqbwe.com	DGA-based Dridex domain
IP Address	185.234.247.119	Known Dridex C2
File Name	oiioiashdqbwe.rar	Initial dropper
File Name	bot_update.dll	C2 payload
URI Path	/update/gate.php	C2 communication endpoint
User-Agent	Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1)	Fake user agent used by malware

6. Impact Assessment

Category	Description
System Impact	Credential theft, process injection, persistence through registry keys.
Network Impact	Unauthorized outbound C2 traffic, DNS beaconing, possible data exfiltration.
Business Risk	Potential compromise of financial or user data, network infiltration risk.

7. Recommended Remediation

Immediate Containment

- Isolate infected host from the network.
- Block malicious IPs/domains in firewall and DNS filters.
- Terminate any suspicious processes (bot_update.dll).

Eradication

- Delete malicious registry keys.
- Clean residual files (.rar, .dll) from system directories.
- Use updated antivirus/malware removal tools.

Recovery

- Restore system from clean backup.
- Reset all credentials and enforce MFA.
- Patch OS and applications.

Prevention

- Implement user awareness training.
- Enable email sandboxing and content disarm features.
- Deploy IDS/IPS rules to detect Dridex-like traffic.

8. Impact Assessment (CIA Triad)

Security Principle	Impact	Assessment Summary
Confidentiality	High	Sensitive user credentials and system data may have been exfiltrated through HTTP POST requests and DNS tunneling activity. Malware such as Dridex is known to harvest banking and credential data.
Integrity	Medium	Possible modification of system files and registry entries to maintain persistence. No verified file tampering detected, but threat behavior suggests data manipulation risk.
Availability	Medium	System performance degradation observed. Malware may attempt to disable antivirus and security tools, affecting service availability. No system crash reported at this stage.

9. Immediate Actions Required

- Isolate the affected host from the corporate network to prevent further lateral movement.
- Block malicious IP addresses and domains identified in the IOC list at the firewall and proxy level.
- Quarantine and analyze the .rar payload (oiioiashdqbwe.rar) using sandboxing.
- Reset all user credentials potentially exposed during infection.
- Perform a full antivirus and memory scan on all potentially affected endpoints.
- Check for persistence mechanisms (registry autoruns, scheduled tasks, PowerShell scripts).
- Notify the SOC team and management for incident escalation and reporting compliance.

10. Recommended Security Improvements (Technical Controls – within 30 Days)

Area	Technical Control	Implementation Timeline
Network Security	Deploy <i>Intrusion Detection & Prevention Systems (IDS/IPS)</i> and enable deep packet inspection for HTTP/DNS anomalies.	0–30 days
Endpoint Protection	Upgrade endpoint security with <i>EDR/XDR</i> capabilities for behavioral detection and rollback.	0–30 days
Web Filtering	Enforce <i>URL categorization</i> and block suspicious or newly registered domains.	0–30 days
Patch Management	Ensure all browsers and OS components are updated; disable legacy IE engines.	0–30 days
Email Security	Strengthen email filtering with sandboxing for attachments and links.	0–30 days
Logging & Monitoring	Centralize logs in <i>SIEM</i> and enable threat correlation alerts for similar patterns.	0–30 days

11. Process Improvements

- Establish a standardized incident response **playbook** for malware-infected endpoints.
- Implement threat hunting routines to identify dormant infections or lateral movement.
- Enhance user awareness training focused on phishing and social engineering.
- Review and test backup procedures to ensure recoverability after a ransomware or data loss event.
- Conduct periodic tabletop exercises to validate SOC and IR readiness.

12. Lessons Learned

Aspect	Observation	Action Point
What Failed	Lack of proactive monitoring allowed C2 traffic to persist for hours undetected.	Implement continuous network traffic monitoring and anomaly detection.
What Worked	Network capture and retrospective forensic analysis successfully identified malicious communication and payload indicators.	Standardize forensic capture on all critical network segments.
Key Takeaways	Malware leveraged legitimate protocols (HTTP/DNS) and	Correlate behavioral and signature-based

Aspect	Observation	Action Point
	outdated User-Agents to disguise traffic.	detection to reduce false negatives.

13. Conclusion

What We Know

The infection originated from HTTP traffic downloading a suspicious .rar archive from an external IP.

Dridex-related communication occurred over DNS and HTTP with outdated browser headers.

The host was beaconing to known malicious C2 domains.

What We Need

Confirm whether any credentials or sensitive data were exfiltrated.

Conduct host-based memory and persistence analysis.

Correlate logs across the organization to detect lateral movement.

15. Report Status

Status: Final (Forensic Analysis Completed)

Version: 1.0

Date Completed: 2025-10-29

Analyst: David Emmanuel

16. Classification

Level	Description
Confidential	Contains sensitive threat intelligence, forensic indicators, and internal IP data. For authorized recipients only.

17. Distribution

Recipient	Department	Access Level
SOC Team	Security Operations	Full
Incident Response	Forensics Unit	Full
Management	Executive Review	Summary
External Partner (if any)	Threat Intelligence Sharing	Redacted

14. Contact Information

Role	Name / Contact
Lead Forensic Analyst	David Emmanuel
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Phone	+44 9045543441
Office Address	DELAP WARE INC, Manchester, UK